

# Western MA Per Pupil Expenditure Report

With selected districts for comparison

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# Appendix A. Data Sources & Calculation Methodology

## 1. Data sources

### Expenditure dataset

Massachusetts Department of Elementary and Secondary Education (DESE)

District Expenditures by Spending Category

Last updated: August 12, 2025

URL: <https://educationtocareer.data.mass.gov/Finance-and-Budget/District-Expenditures-by-Spending-Category/er3w-dyti/>

This dataset provides:

- Expenditures Per Pupil by category for all Massachusetts school districts
- Student enrollment (In-District FTE, Out-of-District FTE, Total FTE)
- Annual End-of-year-report (EOYR) data from FY2009 to present

### Chapter 70 District Profiles

Massachusetts Department of Elementary and Secondary Education (DESE)

Chapter 70 District Profiles

Retrieved: September 2025

URL: <https://www.doe.mass.edu/finance/chapter70/default.html>

From the Ch70 website: "Chapter 70 District Profiles: The on-line Chapter 70 database shows, for each school district, yearly spending and state aid totals in comparison to the foundation budget. Trend data is available for each year going back to FY1993."

This dataset provides:

- Chapter 70 Aid (c70aid): State aid received by each district
- Required Net School Spending (rqdnss2): Minimum spending required by state law
- Actual Net School Spending (actualNSS): Total district spending on education
- Foundation Enrollment (distfoundenro): District foundation enrollment used for Ch70 funding calculations
- Data reported annually each Fall from FY1993 to present

### District classifications and geography

MassGIS Data: Public School Districts

Polygon data layers depicting the boundaries of public school districts in the Commonwealth of Massachusetts.

School type classifications (Traditional, Regional, Charter, etc.)

Last updated: June 2019

URL: <https://www.mass.gov/info-details/massgis-data-public-school-districts>

MassGIS Data: MA Executive Office of Health & Human Services Regions

Districts classified by region based on EOHHS regional designations.

Last updated: April 2022

URL: <https://www.mass.gov/info-details/massgis-data-ma-executive-office-of-health-human-services-regions>

## 2. Key definitions & calculations

### Per-pupil expenditure (PPE)

PPE is reported by the DESE EOYR. It is calculated with In-District FTE enrollment in the denominator. Per DESE's Researcher's Guide: "Out-of-district costs cannot be properly reported as per-pupil expenditure because tuition costs vary greatly depending on the reason for going out of district."

PPE includes 10 subcategories in the EOYR:

- Administration
- Guidance, Counseling and Testing
- Instructional Leadership

- Instructional Materials, Equipment and Technology
- Insurance, Retirement Programs and Other
- Operations and Maintenance
- Other Teaching Services
- Professional Development
- Pupil Services
- Teachers

This report groups two subcategories in 'Other':

- Instructional Materials, Equipment and Technology
- Professional Development

## Chapter 70 and Net school spending (NSS)

To understand funding sources, we decompose total spending into three stacked components:

- Chapter 70 Aid (\$/pupil):  $\text{State funding} = \text{c70aid} / \text{foundation enrollment}$
- Required local contribution (\$/pupil) =  $\max(0, \text{rqdnss2} - \text{c70aid}) / \text{foundation enrollment}$
- Actual local contribution above required (\$/pupil) =  $(\text{actualNSS} - \text{rqdnss2}) / \text{foundation enrollment}$

Total NSS per foundation pupil = Sum of all three components =  $\text{actualNSS} / \text{foundation enrollment}$

## Note about enrollment denominators

Chapter 70 aid and net school spending calculations use foundation enrollment for per-pupil figures, while PPE totals and expense categories use in-district FTE enrollment. These metrics come from different DESE reporting touchpoints:

- Foundation enrollment (Chapter 70 profile): All students residing in the district on October 1st of the prior year for whom the district is financially responsible.
- In-district FTE (End-of-year-report (EOYR)): Full-year enrollment calculated from total membership days for students either residing in the district, or choiced or tuitioned into the district.

This difference complicates direct comparison between Chapter 70 and expense analyses, but remains valid since each district uses the same denominator type within each metric. The comparison is not apples-to-oranges, but a picture of apples in the morning compared with a picture of apples in the evening. For more information, see the [DESE Researcher's Guide](https://www.doe.mass.edu/research/researchers.html): <https://www.doe.mass.edu/research/researchers.html>

## Compound Annual Growth Rate (CAGR)

CAGR measures the smoothed annual growth rate over multiple years — answering "what steady annual rate would produce the same start-to-end result?"

Formula:  $\text{CAGR} = (\text{End\_Value} / \text{Start\_Value})^{(1 / \text{Years})} - 1$

Example: If expenditures grow from \$10,000 to \$15,000 over 10 years:

$\text{CAGR} = (\$15,000 / \$10,000)^{(1/10)} - 1 = 4.14\% \text{ per year}$

## Weighted average

For cohort aggregates, we use enrollment-weighted averages to ensure larger districts appropriately influence the cohort average:

Formula:  $\text{Weighted\_PPE} = \Sigma(\text{District\_PPE} \times \text{District\_FTE}) / \Sigma(\text{District\_FTE})$

## 3. Enrollment-based cohort methodology

Districts are grouped by FY2024 In-District FTE enrollment using quartile analysis:

### Cohort boundaries (FY2024)

Based on 59 Western MA traditional districts:

- Q1 (25th percentile): 154.7 FTE → rounded to 200
- Median (50th percentile): 792.0 FTE → rounded to 800
- Q3 (75th percentile): 1,597.6 FTE → rounded to 1,600
- Outlier threshold: Fixed at 10,000 FTE

Final cohorts:

- **Tiny (0-200 FTE):** 17 districts (Cohort 1: below Q1) **Member districts:** conway, erving, farmington river reg, florida, hancock, hawlemont, leverett, new salem-wendell, pelham, petersham, richmond, rowe, savoy, shutesbury, sunderland, and 2 others
- **Small (201-800 FTE):** 13 districts (Cohort 2: Q1 to median) **Member districts:** clarksburg, deerfield, frontier, gateway, granby, hadley, hatfield, lee, lenox, orange, pioneer valley, ralph c mahar, southern berkshire
- **Medium (801-1600 FTE):** 15 districts (Cohort 3: median to Q3) **Member districts:** amherst, amherst-pelham, berkshire hills, central berkshire, easthampton, gill-montague, greenfield, hoosac valley regional, mohawk trail, monson, mount greylock, north adams, palmer, southwick-tolland-granville regional school district, ware
- **Large (1601-10K FTE):** 14 districts (Cohort 4: above Q3) **Member districts:** agawam, athol-royalston, belchertown, chicopee, east longmeadow, hampden-wilbraham, holyoke, longmeadow, ludlow, northampton, pittsfield, south hadley, west springfield, westfield
- **Outliers (Springfield >10K FTE):** 1 district (Cohort 5: statistical outlier, analyzed separately) **Member districts:** springfield

## 4. Comparison shading methodology

To visualize meaningful differences while filtering statistical noise, we use graduated color shading based on two independent tests:

### Colors

- Orange: District exceeds baseline
- Blue: District below baseline
- White: Within threshold (not meaningfully different)

### Test 1: Dollar amounts (2024 values)

- Threshold:  $\geq 5.0\%$  difference from baseline
- Calculation:  $(\text{District} - \text{Baseline}) / \text{Baseline}$
- Shading intensity bins: 5%, 10%, 15%, 20%+

### Test 2: Growth rates (CAGR)

- Threshold:  $\geq 0.5$  percentage points (pp) difference from baseline
- Calculation:  $\text{District\_CAGR} - \text{Baseline\_CAGR}$
- Shading intensity bins: 0pp, 1pp, 2pp, 2pp+

### Statistical justification for thresholds

Analysis of all 59 Western MA districts revealed different natural variation by metric:

- PPE variation: Mean = \$24,237, SD = \$5,462, CV = 22.5%
- CAGR variation: Mean = 6.00pp, SD = 3.24pp, CV = 54.0%

The 5.0%/0.5pp thresholds balance statistical rigor with practical communication:

- 5.0% for PPE: Flags differences  $\geq \$1,212$  at mean ( $\sim 0.22$  SD)
- 0.5pp for CAGR: Flags differences  $\geq 0.5$ pp ( $\sim 0.15$  SD)
- Balance ratio:  $0.7\times$  (appropriately sensitive to compound growth effects)

This approach creates a visual hierarchy:

- No shading: Districts statistically similar to peers
- Lighter shading: Notable difference
- Darker shading: Exceptional difference requiring attention

These thresholds reflect the different ways these metrics behave. PPE and enrollment are dollar amounts and headcounts — a

5.0% difference is substantial and visible in annual budgets. But CAGR measures rates of change, and small differences in growth rates compound dramatically over time. Consider a district with \$20,000 PPE:

- Growing at 4.0% annually reaches \$36,019 after 15 years
- Growing at 4.5% annually reaches \$38,318 after 15 years
- That 0.5pp difference produces a \$2,299 gap (11.5% more total growth)

For a 300-student district, that's \$689,700 in additional annual spending by year 15. For municipalities managing long-term budgets and multi-year labor contracts, these trajectories matter—even when year-to-year changes seem modest.

The tighter CAGR threshold (0.5pp vs. 5%) also reflects statistical reality: growth rates vary more widely across districts (CV = 54%) than PPE levels do (CV = 22.5%). A 0.5pp threshold flags districts with growth trajectories meaningfully different from their cohort without overcounting normal variation. By using a more sensitive threshold for CAGR, the analysis captures the real-world policy significance of compound growth differences while maintaining appropriate sensitivity for dollar and enrollment comparisons.

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## Appendix B. Data Tables

### All data values used in plots

This appendix contains the underlying data tables for all districts and regions shown in the report. Each table shows PPE by category (in \$/in-district FTE pupil), FTE enrollment counts, and NSS/Ch70 funding components (in \$/foundation enrollment pupil) across all available years.

### Data: All Western MA Traditional Districts: Tiny (0-200 FTE)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

#### PPE by Category (\$/in-district FTE pupil)

Table 1

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$577    | \$571    | \$623    | \$643    | \$647    | \$665    | \$676    | \$709    | \$780    | \$579    | \$673    | \$738    | \$1,380  | \$1,042  | \$1,304  | \$1,057  |
| Guidance, Counseling and Testing | \$345    | \$365    | \$375    | \$402    | \$419    | \$462    | \$475    | \$454    | \$445    | \$448    | \$459    | \$488    | \$577    | \$663    | \$621    | \$674    |
| Administration                   | \$770    | \$808    | \$814    | \$853    | \$915    | \$851    | \$774    | \$819    | \$863    | \$869    | \$770    | \$833    | \$950    | \$1,003  | \$1,114  | \$1,126  |
| Instructional Leadership         | \$1,042  | \$1,096  | \$1,125  | \$1,225  | \$1,269  | \$1,316  | \$1,496  | \$1,584  | \$1,510  | \$1,524  | \$1,543  | \$1,589  | \$1,817  | \$1,733  | \$1,777  | \$1,991  |
| Operations and Maintenance       | \$1,374  | \$1,312  | \$1,460  | \$1,416  | \$1,359  | \$1,558  | \$1,554  | \$1,452  | \$1,433  | \$1,468  | \$1,811  | \$1,670  | \$2,060  | \$2,389  | \$2,232  | \$2,652  |
| Other Teaching Services          | \$1,899  | \$1,926  | \$1,987  | \$2,112  | \$2,412  | \$2,378  | \$2,494  | \$2,635  | \$2,673  | \$2,707  | \$3,004  | \$3,115  | \$3,322  | \$3,677  | \$3,638  | \$3,757  |
| Pupil Services                   | \$1,324  | \$1,427  | \$1,426  | \$1,497  | \$1,605  | \$1,599  | \$1,670  | \$1,753  | \$1,846  | \$1,813  | \$1,864  | \$1,716  | \$2,119  | \$2,010  | \$2,334  | \$2,732  |
| Insurance, Retirement and Other  | \$2,258  | \$2,405  | \$2,595  | \$2,737  | \$2,926  | \$3,148  | \$3,125  | \$3,143  | \$3,905  | \$3,390  | \$3,662  | \$3,769  | \$4,357  | \$4,048  | \$4,506  | \$5,009  |
| Teachers                         | \$5,148  | \$5,382  | \$5,545  | \$5,277  | \$5,430  | \$5,906  | \$6,187  | \$6,222  | \$6,245  | \$6,520  | \$6,792  | \$6,982  | \$8,015  | \$7,858  | \$7,808  | \$8,507  |
| Total                            | \$14,738 | \$15,291 | \$15,950 | \$16,161 | \$16,981 | \$17,882 | \$18,452 | \$18,770 | \$19,700 | \$19,319 | \$20,577 | \$20,900 | \$24,596 | \$24,423 | \$25,334 | \$27,504 |

#### FTE Enrollment

Table 2

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 131  | 127  | 125  | 125  | 123  | 120  | 122  | 116  | 121  | 123  | 122  | 122  | 107  | 115  | 114  | 113  |
| Foundation Enrollment      | 123  | 120  | 116  | 109  | 109  | 108  | 107  | 111  | 108  | 107  | 105  | 102  | 101  | 91   | 93   | 94   |
| Out-of-District FTE Pupils | 41   | 41   | 41   | 37   | 40   | 38   | 37   | 39   | 38   | 36   | 35   | 33   | 31   | 34   | 34   | 32   |

#### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 3

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$3K  | \$3K  | \$3K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  |
| Req NSS (minus Ch70)       | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$9K  | \$9K  | \$10K |
| Actual NSS (minus Req NSS) | \$5K  | \$4K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$8K  | \$8K  | \$10K | \$10K | \$10K | \$12K | \$12K | \$12K |
| Total NSS                  | \$13K | \$14K | \$15K | \$16K | \$17K | \$17K | \$17K | \$18K | \$20K | \$20K | \$22K | \$22K | \$23K | \$26K | \$26K | \$27K |

## Data: All Western MA Traditional Districts: Small (201-800 FTE)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 4

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$558    | \$505    | \$540    | \$533    | \$585    | \$515    | \$503    | \$583    | \$539    | \$596    | \$558    | \$512    | \$915    | \$777    | \$1,064  | \$1,150  |
| Guidance, Counseling and Testing | \$392    | \$422    | \$418    | \$456    | \$483    | \$482    | \$501    | \$564    | \$588    | \$600    | \$669    | \$662    | \$807    | \$864    | \$896    | \$961    |
| Administration                   | \$523    | \$515    | \$567    | \$601    | \$648    | \$665    | \$701    | \$739    | \$999    | \$752    | \$788    | \$802    | \$854    | \$942    | \$948    | \$976    |
| Instructional Leadership         | \$811    | \$805    | \$838    | \$824    | \$909    | \$947    | \$967    | \$1,024  | \$1,028  | \$1,042  | \$1,016  | \$1,097  | \$1,188  | \$1,260  | \$1,336  | \$1,463  |
| Operations and Maintenance       | \$1,108  | \$1,056  | \$1,043  | \$1,101  | \$1,198  | \$1,264  | \$1,295  | \$1,338  | \$1,254  | \$1,285  | \$1,473  | \$1,422  | \$1,676  | \$1,878  | \$2,176  | \$2,454  |
| Other Teaching Services          | \$1,061  | \$1,089  | \$1,193  | \$1,151  | \$1,289  | \$1,311  | \$1,353  | \$1,446  | \$1,575  | \$1,718  | \$1,799  | \$1,806  | \$1,779  | \$1,899  | \$2,151  | \$2,402  |
| Pupil Services                   | \$1,421  | \$1,366  | \$1,441  | \$1,537  | \$1,733  | \$1,739  | \$1,714  | \$1,913  | \$1,850  | \$1,870  | \$1,974  | \$2,007  | \$2,097  | \$2,569  | \$2,760  | \$3,122  |
| Insurance, Retirement and Other  | \$2,128  | \$2,107  | \$2,230  | \$2,317  | \$2,486  | \$2,568  | \$2,711  | \$2,955  | \$3,219  | \$3,345  | \$3,362  | \$3,420  | \$3,734  | \$3,801  | \$3,969  | \$4,254  |
| Teachers                         | \$4,522  | \$4,588  | \$4,679  | \$4,870  | \$5,353  | \$5,423  | \$5,557  | \$5,771  | \$6,026  | \$6,075  | \$6,152  | \$6,450  | \$6,922  | \$7,338  | \$7,596  | \$7,838  |
| Total                            | \$12,523 | \$12,453 | \$12,949 | \$13,388 | \$14,684 | \$14,916 | \$15,302 | \$16,333 | \$17,077 | \$17,283 | \$17,791 | \$18,177 | \$19,972 | \$21,330 | \$22,896 | \$24,622 |

### FTE Enrollment

Table 5

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 775  | 779  | 763  | 742  | 696  | 680  | 661  | 644  | 628  | 614  | 617  | 601  | 560  | 556  | 553  | 542  |
| Foundation Enrollment      | 695  | 676  | 666  | 646  | 628  | 603  | 590  | 559  | 542  | 536  | 528  | 526  | 522  | 494  | 484  | 481  |
| Out-of-District FTE Pupils | 75   | 78   | 73   | 73   | 80   | 82   | 86   | 82   | 85   | 88   | 90   | 88   | 85   | 87   | 87   | 84   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 6

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  |
| Req NSS (minus Ch70)       | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  | \$9K  | \$9K  |
| Actual NSS (minus Req NSS) | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$3K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  |
| Total NSS                  | \$11K | \$11K | \$11K | \$12K | \$13K | \$14K | \$14K | \$15K | \$16K | \$16K | \$17K | \$17K | \$18K | \$19K | \$21K | \$21K |

## Data: All Western MA Traditional Districts: Medium (801-1600 FTE)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 7

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$461    | \$484    | \$558    | \$494    | \$561    | \$552    | \$524    | \$573    | \$586    | \$534    | \$644    | \$607    | \$930    | \$1,064  | \$1,055  | \$1,200  |
| Guidance, Counseling and Testing | \$373    | \$385    | \$391    | \$418    | \$435    | \$457    | \$470    | \$479    | \$522    | \$536    | \$545    | \$601    | \$677    | \$736    | \$805    | \$826    |
| Administration                   | \$494    | \$521    | \$512    | \$533    | \$575    | \$576    | \$614    | \$653    | \$678    | \$711    | \$653    | \$700    | \$747    | \$785    | \$835    | \$911    |
| Instructional Leadership         | \$893    | \$902    | \$885    | \$1,039  | \$1,062  | \$1,091  | \$1,113  | \$1,169  | \$1,150  | \$1,071  | \$1,063  | \$1,113  | \$1,191  | \$1,263  | \$1,373  | \$1,573  |
| Operations and Maintenance       | \$1,150  | \$1,085  | \$1,155  | \$1,095  | \$1,146  | \$1,208  | \$1,233  | \$1,218  | \$1,228  | \$1,203  | \$1,326  | \$1,383  | \$1,564  | \$1,663  | \$1,836  | \$1,953  |
| Other Teaching Services          | \$1,014  | \$1,078  | \$1,100  | \$1,135  | \$1,235  | \$1,317  | \$1,362  | \$1,400  | \$1,444  | \$1,596  | \$1,681  | \$1,729  | \$1,936  | \$2,180  | \$2,311  | \$2,395  |
| Pupil Services                   | \$1,351  | \$1,309  | \$1,366  | \$1,345  | \$1,466  | \$1,527  | \$1,626  | \$1,705  | \$1,686  | \$1,869  | \$1,919  | \$1,803  | \$1,957  | \$2,231  | \$2,548  | \$2,755  |
| Insurance, Retirement and Other  | \$2,483  | \$2,574  | \$2,743  | \$2,707  | \$2,707  | \$2,943  | \$3,074  | \$3,251  | \$3,266  | \$3,622  | \$3,631  | \$3,789  | \$4,169  | \$4,166  | \$4,480  | \$4,740  |
| Teachers                         | \$4,813  | \$4,890  | \$5,055  | \$5,048  | \$5,152  | \$5,357  | \$5,513  | \$5,578  | \$5,674  | \$5,960  | \$6,251  | \$6,385  | \$6,758  | \$6,773  | \$7,182  | \$7,431  |
| Total                            | \$13,032 | \$13,227 | \$13,766 | \$13,815 | \$14,340 | \$15,028 | \$15,529 | \$16,025 | \$16,234 | \$17,103 | \$17,713 | \$18,110 | \$19,930 | \$20,859 | \$22,426 | \$23,785 |

### FTE Enrollment

Table 8

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 1,466 | 1,434 | 1,416 | 1,402 | 1,388 | 1,334 | 1,306 | 1,277 | 1,268 | 1,247 | 1,268 | 1,249 | 1,160 | 1,175 | 1,158 | 1,144 |
| Foundation Enrollment      | 1,622 | 1,596 | 1,567 | 1,528 | 1,509 | 1,507 | 1,472 | 1,451 | 1,422 | 1,399 | 1,381 | 1,376 | 1,361 | 1,289 | 1,283 | 1,247 |
| Out-of-District FTE Pupils | 154   | 161   | 164   | 164   | 166   | 167   | 176   | 185   | 187   | 194   | 201   | 198   | 190   | 191   | 194   | 194   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 9

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  |
| Req NSS (minus Ch70)       | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$8K  |
| Actual NSS (minus Req NSS) | \$2K  | \$1K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$4K  | \$4K  |
| Total NSS                  | \$11K | \$11K | \$11K | \$12K | \$12K | \$13K | \$13K | \$14K | \$14K | \$15K | \$16K | \$16K | \$16K | \$17K | \$18K | \$20K |



## Data: All Western MA Traditional Districts: Large (1601-10K FTE)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 10

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$513    | \$648    | \$614    | \$611    | \$654    | \$631    | \$610    | \$644    | \$664    | \$667    | \$722    | \$716    | \$972    | \$1,075  | \$1,331  | \$1,348  |
| Guidance, Counseling and Testing | \$397    | \$394    | \$416    | \$407    | \$423    | \$443    | \$452    | \$454    | \$454    | \$483    | \$511    | \$537    | \$597    | \$664    | \$741    | \$740    |
| Administration                   | \$392    | \$415    | \$426    | \$437    | \$441    | \$462    | \$484    | \$496    | \$480    | \$494    | \$453    | \$502    | \$537    | \$571    | \$616    | \$649    |
| Instructional Leadership         | \$811    | \$859    | \$865    | \$856    | \$837    | \$886    | \$915    | \$932    | \$956    | \$940    | \$998    | \$1,028  | \$1,060  | \$1,153  | \$1,195  | \$1,231  |
| Operations and Maintenance       | \$1,014  | \$993    | \$1,009  | \$986    | \$984    | \$1,023  | \$1,093  | \$1,098  | \$1,111  | \$1,152  | \$1,244  | \$1,238  | \$1,401  | \$1,450  | \$1,685  | \$1,683  |
| Other Teaching Services          | \$990    | \$1,037  | \$1,103  | \$1,105  | \$1,177  | \$1,228  | \$1,290  | \$1,365  | \$1,383  | \$1,395  | \$1,407  | \$1,429  | \$1,561  | \$1,717  | \$1,951  | \$2,021  |
| Pupil Services                   | \$1,135  | \$1,141  | \$1,206  | \$1,205  | \$1,274  | \$1,354  | \$1,396  | \$1,472  | \$1,495  | \$1,586  | \$1,659  | \$1,591  | \$1,603  | \$1,989  | \$2,213  | \$2,408  |
| Insurance, Retirement and Other  | \$1,911  | \$1,943  | \$2,045  | \$2,123  | \$2,115  | \$2,282  | \$2,366  | \$2,419  | \$2,543  | \$2,652  | \$2,773  | \$2,855  | \$3,037  | \$3,212  | \$3,364  | \$3,469  |
| Teachers                         | \$4,613  | \$4,604  | \$4,707  | \$4,868  | \$4,990  | \$5,067  | \$5,194  | \$5,393  | \$5,482  | \$5,569  | \$5,863  | \$6,031  | \$6,461  | \$6,780  | \$7,001  | \$7,292  |
| Total                            | \$11,775 | \$12,034 | \$12,391 | \$12,597 | \$12,895 | \$13,376 | \$13,801 | \$14,273 | \$14,568 | \$14,937 | \$15,629 | \$15,927 | \$17,229 | \$18,612 | \$20,097 | \$20,841 |

### FTE Enrollment

Table 11

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 4,001 | 3,953 | 3,935 | 3,852 | 3,812 | 3,760 | 3,721 | 3,674 | 3,690 | 3,680 | 3,616 | 3,598 | 3,397 | 3,422 | 3,384 | 3,379 |
| Foundation Enrollment      | 4,432 | 4,382 | 4,347 | 4,325 | 4,280 | 4,230 | 4,191 | 4,163 | 4,093 | 4,077 | 4,037 | 4,013 | 3,975 | 3,823 | 3,787 | 3,752 |
| Out-of-District FTE Pupils | 215   | 227   | 244   | 253   | 256   | 272   | 293   | 309   | 316   | 322   | 341   | 345   | 344   | 341   | 354   | 357   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 12

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$6K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  | \$9K  | \$10K |
| Req NSS (minus Ch70)       | \$3K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  |
| Actual NSS (minus Req NSS) | \$900 | \$708 | \$901 | \$905 | \$944 | \$1K  | \$1K  | \$1K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  |
| Total NSS                  | \$10K | \$10K | \$10K | \$11K | \$11K | \$12K | \$12K | \$12K | \$13K | \$13K | \$14K | \$14K | \$15K | \$16K | \$17K | \$18K |

## Data: All Western MA Traditional Districts: Outliers (Springfield at 23,723 FTE in 2024)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 13

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$1,143  | \$1,337  | \$2,068  | \$1,143  | \$1,424  | \$1,734  | \$1,241  | \$1,376  | \$1,286  | \$1,371  | \$1,673  | \$1,765  | \$2,487  | \$2,260  | \$2,923  | \$3,071  |
| Guidance, Counseling and Testing | \$380    | \$362    | \$427    | \$388    | \$400    | \$405    | \$408    | \$389    | \$391    | \$438    | \$469    | \$483    | \$513    | \$522    | \$621    | \$697    |
| Administration                   | \$533    | \$654    | \$446    | \$645    | \$550    | \$677    | \$596    | \$580    | \$507    | \$490    | \$416    | \$407    | \$441    | \$501    | \$599    | \$609    |
| Instructional Leadership         | \$719    | \$778    | \$892    | \$1,173  | \$1,257  | \$1,245  | \$1,208  | \$1,282  | \$1,280  | \$1,060  | \$983    | \$1,090  | \$1,103  | \$1,402  | \$1,534  | \$1,571  |
| Operations and Maintenance       | \$1,150  | \$1,081  | \$1,262  | \$1,176  | \$1,132  | \$997    | \$1,121  | \$1,027  | \$1,063  | \$1,156  | \$1,337  | \$1,093  | \$1,534  | \$1,536  | \$3,453  | \$3,806  |
| Other Teaching Services          | \$1,143  | \$1,002  | \$1,146  | \$1,130  | \$1,051  | \$1,086  | \$1,150  | \$1,135  | \$1,070  | \$1,036  | \$1,357  | \$1,390  | \$1,337  | \$1,518  | \$1,776  | \$2,014  |
| Pupil Services                   | \$1,651  | \$1,511  | \$1,729  | \$1,965  | \$1,783  | \$1,852  | \$1,985  | \$2,078  | \$2,222  | \$2,407  | \$2,499  | \$2,148  | \$2,077  | \$3,080  | \$3,054  | \$3,204  |
| Insurance, Retirement and Other  | \$2,050  | \$2,226  | \$2,280  | \$2,474  | \$2,437  | \$2,496  | \$2,467  | \$2,566  | \$2,562  | \$2,635  | \$2,765  | \$3,001  | \$3,217  | \$3,461  | \$3,852  | \$3,906  |
| Teachers                         | \$5,839  | \$4,859  | \$4,750  | \$4,711  | \$4,848  | \$4,823  | \$4,963  | \$5,196  | \$4,962  | \$5,274  | \$5,510  | \$5,800  | \$6,442  | \$7,095  | \$7,561  | \$7,737  |
| Total                            | \$14,608 | \$13,810 | \$15,000 | \$14,805 | \$14,882 | \$15,315 | \$15,139 | \$15,629 | \$15,343 | \$15,867 | \$17,009 | \$17,177 | \$19,151 | \$21,375 | \$25,373 | \$26,615 |

### FTE Enrollment

Table 14

| FTE Series                 | 2009   | 2010   | 2011   | 2012   | 2013   | 2014   | 2015   | 2016   | 2017   | 2018   | 2019   | 2020   | 2021   | 2022   | 2023   | 2024   |
|----------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| In-District FTE Pupils     | 25,137 | 24,987 | 24,935 | 24,521 | 24,747 | 25,249 | 25,101 | 25,009 | 25,629 | 25,776 | 25,294 | 24,906 | 24,079 | 24,040 | 23,763 | 23,723 |
| Foundation Enrollment      | 28,235 | 28,343 | 28,305 | 28,226 | 27,951 | 28,434 | 28,755 | 28,970 | 29,109 | 29,533 | 29,670 | 29,645 | 29,551 | 29,144 | 28,613 | 28,326 |
| Out-of-District FTE Pupils | 3,571  | 3,775  | 3,719  | 3,695  | 3,817  | 3,613  | 3,974  | 4,406  | 4,596  | 4,812  | 5,150  | 5,363  | 5,559  | 5,500  | 5,652  | 5,745  |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 15

| Component                  | 2009   | 2010   | 2011  | 2012  | 2013   | 2014  | 2015  | 2016  | 2017  | 2018   | 2019  | 2020   | 2021  | 2022  | 2023  | 2024  |
|----------------------------|--------|--------|-------|-------|--------|-------|-------|-------|-------|--------|-------|--------|-------|-------|-------|-------|
| Ch70 Aid                   | \$9K   | \$9K   | \$9K  | \$10K | \$10K  | \$10K | \$10K | \$11K | \$11K | \$11K  | \$12K | \$13K  | \$13K | \$14K | \$15K | \$17K |
| Req NSS (minus Ch70)       | \$247  | \$1K   | \$2K  | \$1K  | \$1K   | \$1K  | \$1K  | \$1K  | \$1K  | \$1K   | \$1K  | \$1K   | \$2K  | \$1K  | \$2K  | \$2K  |
| Actual NSS (minus Req NSS) | \$-150 | \$-384 | \$-35 | \$-84 | \$-144 | \$0   | \$2   | \$2   | \$-0  | \$-106 | \$-0  | \$-499 | \$0   | \$0   | \$-0  | \$-0  |
| Total NSS                  | \$9K   | \$10K  | \$11K | \$11K | \$11K  | \$12K | \$12K | \$12K | \$12K | \$12K  | \$13K | \$13K  | \$15K | \$15K | \$17K | \$19K |

## Data: All Western MA Traditional Districts: All Western MA (excl. Springfield)

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 16

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$507    | \$590    | \$593    | \$575    | \$624    | \$600    | \$580    | \$622    | \$636    | \$627    | \$686    | \$671    | \$967    | \$1,043  | \$1,240  | \$1,287  |
| Guidance, Counseling and Testing | \$389    | \$394    | \$409    | \$415    | \$432    | \$451    | \$462    | \$471    | \$483    | \$506    | \$534    | \$564    | \$636    | \$700    | \$768    | \$780    |
| Administration                   | \$440    | \$461    | \$471    | \$488    | \$507    | \$520    | \$544    | \$566    | \$589    | \$580    | \$542    | \$587    | \$628    | \$669    | \$713    | \$754    |
| Instructional Leadership         | \$836    | \$869    | \$873    | \$906    | \$910    | \$951    | \$981    | \$1,013  | \$1,023  | \$996    | \$1,029  | \$1,070  | \$1,122  | \$1,205  | \$1,266  | \$1,353  |
| Operations and Maintenance       | \$1,066  | \$1,030  | \$1,058  | \$1,035  | \$1,055  | \$1,105  | \$1,158  | \$1,160  | \$1,161  | \$1,185  | \$1,301  | \$1,302  | \$1,482  | \$1,566  | \$1,783  | \$1,845  |
| Other Teaching Services          | \$1,025  | \$1,073  | \$1,133  | \$1,141  | \$1,232  | \$1,284  | \$1,342  | \$1,413  | \$1,449  | \$1,508  | \$1,553  | \$1,582  | \$1,714  | \$1,895  | \$2,100  | \$2,190  |
| Pupil Services                   | \$1,224  | \$1,214  | \$1,276  | \$1,283  | \$1,377  | \$1,442  | \$1,491  | \$1,580  | \$1,584  | \$1,686  | \$1,758  | \$1,685  | \$1,747  | \$2,103  | \$2,348  | \$2,566  |
| Insurance, Retirement and Other  | \$2,083  | \$2,124  | \$2,247  | \$2,301  | \$2,318  | \$2,491  | \$2,589  | \$2,689  | \$2,816  | \$2,966  | \$3,060  | \$3,157  | \$3,404  | \$3,516  | \$3,715  | \$3,881  |
| Teachers                         | \$4,664  | \$4,689  | \$4,806  | \$4,921  | \$5,078  | \$5,194  | \$5,331  | \$5,496  | \$5,601  | \$5,735  | \$6,009  | \$6,181  | \$6,614  | \$6,861  | \$7,123  | \$7,409  |
| Total                            | \$12,234 | \$12,444 | \$12,865 | \$13,063 | \$13,532 | \$14,039 | \$14,479 | \$15,010 | \$15,343 | \$15,789 | \$16,472 | \$16,798 | \$18,314 | \$19,558 | \$21,055 | \$22,065 |

### FTE Enrollment

Table 17

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 1,555 | 1,534 | 1,521 | 1,493 | 1,469 | 1,438 | 1,418 | 1,372 | 1,371 | 1,361 | 1,352 | 1,339 | 1,255 | 1,266 | 1,252 | 1,245 |
| Foundation Enrollment      | 1,603 | 1,580 | 1,563 | 1,543 | 1,524 | 1,505 | 1,486 | 1,469 | 1,442 | 1,432 | 1,416 | 1,408 | 1,394 | 1,333 | 1,322 | 1,306 |
| Out-of-District FTE Pupils | 120   | 125   | 129   | 130   | 134   | 138   | 146   | 150   | 152   | 155   | 162   | 161   | 158   | 158   | 162   | 162   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 18

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$8K  | \$8K  | \$9K  |
| Req NSS (minus Ch70)       | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  |
| Actual NSS (minus Req NSS) | \$1K  | \$1K  | \$1K  | \$1K  | \$1K  | \$1K  | \$2K  | \$2K  | \$2K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  |
| Total NSS                  | \$10K | \$11K | \$11K | \$11K | \$12K | \$12K | \$13K | \$13K | \$14K | \$14K | \$15K | \$15K | \$15K | \$17K | \$18K | \$19K |

## Data: Amherst-Pelham

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 19

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$646    | \$534    | \$900    | \$417    | \$459    | \$424    | \$599    | \$860    | \$635    | \$384    | \$610    | \$559    | \$477    | \$527    | \$457    | \$799    |
| Guidance, Counseling and Testing | \$525    | \$580    | \$617    | \$684    | \$652    | \$733    | \$778    | \$798    | \$803    | \$891    | \$925    | \$945    | \$1,001  | \$1,241  | \$1,406  | \$1,465  |
| Administration                   | \$652    | \$664    | \$687    | \$789    | \$987    | \$907    | \$953    | \$926    | \$996    | \$857    | \$759    | \$780    | \$788    | \$859    | \$928    | \$1,277  |
| Instructional Leadership         | \$1,289  | \$1,337  | \$1,209  | \$1,397  | \$1,390  | \$1,545  | \$1,429  | \$1,437  | \$1,541  | \$1,555  | \$1,544  | \$1,501  | \$1,436  | \$1,472  | \$1,644  | \$2,336  |
| Operations and Maintenance       | \$1,459  | \$1,416  | \$1,559  | \$1,468  | \$1,481  | \$1,585  | \$1,473  | \$1,577  | \$1,904  | \$1,678  | \$1,776  | \$1,601  | \$1,705  | \$1,778  | \$1,957  | \$2,221  |
| Other Teaching Services          | \$1,199  | \$1,264  | \$1,313  | \$1,468  | \$1,588  | \$1,739  | \$1,770  | \$1,883  | \$2,014  | \$2,066  | \$2,122  | \$2,214  | \$3,010  | \$3,050  | \$2,968  | \$2,851  |
| Pupil Services                   | \$1,645  | \$1,616  | \$1,588  | \$1,618  | \$1,639  | \$1,764  | \$1,975  | \$2,110  | \$2,128  | \$2,218  | \$2,236  | \$1,957  | \$2,950  | \$2,539  | \$2,771  | \$2,766  |
| Insurance, Retirement and Other  | \$3,305  | \$3,636  | \$3,673  | \$3,811  | \$3,526  | \$3,954  | \$4,086  | \$4,103  | \$4,218  | \$5,006  | \$5,047  | \$5,074  | \$5,237  | \$5,062  | \$5,887  | \$6,248  |
| Teachers                         | \$5,491  | \$5,511  | \$6,254  | \$6,199  | \$6,533  | \$6,765  | \$6,704  | \$6,590  | \$6,927  | \$7,208  | \$7,285  | \$7,155  | \$7,592  | \$7,404  | \$8,158  | \$8,270  |
| Total                            | \$16,211 | \$16,558 | \$17,800 | \$17,851 | \$18,255 | \$19,416 | \$19,767 | \$20,284 | \$21,166 | \$21,863 | \$22,304 | \$21,786 | \$24,196 | \$23,932 | \$26,176 | \$28,233 |

### FTE Enrollment

Table 20

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 1,712 | 1,659 | 1,573 | 1,536 | 1,519 | 1,470 | 1,450 | 1,378 | 1,366 | 1,342 | 1,334 | 1,346 | 1,278 | 1,265 | 1,241 | 1,209 |
| Foundation Enrollment      | 1,801 | 1,763 | 1,726 | 1,647 | 1,610 | 1,581 | 1,529 | 1,532 | 1,496 | 1,490 | 1,433 | 1,399 | 1,399 | 1,330 | 1,310 | 1,274 |
| Out-of-District FTE Pupils | 112   | 129   | 138   | 143   | 138   | 149   | 169   | 190   | 192   | 170   | 158   | 152   | 155   | 145   | 150   | 147   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 21

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  |
| Req NSS (minus Ch70)       | \$4K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$9K  | \$10K | \$10K | \$11K |
| Actual NSS (minus Req NSS) | \$3K  | \$2K  | \$3K  | \$3K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  |
| Total NSS                  | \$13K | \$14K | \$14K | \$16K | \$17K | \$17K | \$18K | \$18K | \$19K | \$19K | \$20K | \$21K | \$21K | \$22K | \$23K | \$25K |

## Data: Amherst

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 22

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$684    | \$547    | \$606    | \$931    | \$1,098  | \$1,098  | \$1,265  | \$1,242  | \$1,119  | \$798    | \$1,092  | \$733    | \$1,563  | \$1,561  | \$1,475  | \$1,328  |
| Guidance, Counseling and Testing | \$396    | \$383    | \$362    | \$411    | \$469    | \$538    | \$545    | \$583    | \$599    | \$631    | \$668    | \$696    | \$830    | \$891    | \$1,093  | \$1,053  |
| Administration                   | \$627    | \$698    | \$735    | \$709    | \$787    | \$809    | \$919    | \$944    | \$981    | \$799    | \$767    | \$827    | \$874    | \$896    | \$981    | \$1,213  |
| Instructional Leadership         | \$1,134  | \$1,079  | \$965    | \$1,032  | \$1,153  | \$1,275  | \$1,175  | \$1,131  | \$1,087  | \$1,231  | \$1,325  | \$1,527  | \$1,750  | \$1,783  | \$1,847  | \$2,498  |
| Operations and Maintenance       | \$1,264  | \$1,271  | \$1,636  | \$1,436  | \$1,399  | \$1,451  | \$1,577  | \$1,530  | \$1,555  | \$1,340  | \$1,646  | \$1,764  | \$1,897  | \$1,874  | \$1,941  | \$2,248  |
| Other Teaching Services          | \$1,762  | \$1,813  | \$1,923  | \$2,345  | \$2,621  | \$2,610  | \$2,840  | \$2,652  | \$2,751  | \$2,973  | \$3,084  | \$3,053  | \$3,908  | \$3,898  | \$3,746  | \$4,612  |
| Pupil Services                   | \$865    | \$1,003  | \$973    | \$975    | \$1,084  | \$1,223  | \$1,257  | \$1,267  | \$1,219  | \$1,278  | \$1,362  | \$1,148  | \$1,205  | \$1,849  | \$1,728  | \$1,965  |
| Insurance, Retirement and Other  | \$3,353  | \$3,610  | \$3,821  | \$3,852  | \$3,941  | \$4,189  | \$4,201  | \$4,295  | \$4,231  | \$4,840  | \$5,212  | \$5,363  | \$5,661  | \$5,561  | \$6,022  | \$6,605  |
| Teachers                         | \$5,944  | \$6,213  | \$6,355  | \$6,963  | \$6,804  | \$6,627  | \$6,841  | \$7,712  | \$7,792  | \$8,091  | \$8,538  | \$8,725  | \$9,528  | \$9,598  | \$9,451  | \$9,745  |
| Total                            | \$16,029 | \$16,617 | \$17,376 | \$18,654 | \$19,356 | \$19,820 | \$20,620 | \$21,356 | \$21,334 | \$21,981 | \$23,694 | \$23,836 | \$27,216 | \$27,911 | \$28,284 | \$31,267 |

### FTE Enrollment

Table 23

| FTE Series                 | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| In-District FTE Pupils     | 1,387 | 1,331 | 1,284 | 1,215 | 1,196 | 1,197 | 1,184 | 1,154 | 1,156 | 1,163 | 1,131 | 1,107 | 1,017 | 1,052 | 1,062 | 1,002 |
| Foundation Enrollment      | 1,398 | 1,405 | 1,356 | 1,279 | 1,269 | 1,227 | 1,205 | 1,192 | 1,199 | 1,136 | 1,106 | 1,134 | 1,097 | 1,045 | 1,065 | 1,055 |
| Out-of-District FTE Pupils | 65    | 72    | 73    | 84    | 90    | 90    | 101   | 113   | 110   | 108   | 124   | 122   | 118   | 123   | 129   | 134   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 24

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  |
| Req NSS (minus Ch70)       | \$4K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$9K  | \$10K | \$11K |
| Actual NSS (minus Req NSS) | \$6K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$7K  | \$8K  | \$8K  | \$9K  | \$10K | \$9K  | \$9K  | \$11K | \$11K | \$11K |
| Total NSS                  | \$15K | \$15K | \$16K | \$17K | \$17K | \$18K | \$19K | \$20K | \$20K | \$22K | \$23K | \$23K | \$24K | \$26K | \$26K | \$28K |

## Data: Leverett

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 25

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$575    | \$602    | \$651    | \$766    | \$616    | \$556    | \$563    | \$529    | \$596    | \$453    | \$503    | \$606    | \$1,041  | \$840    | \$756    | \$608    |
| Guidance, Counseling and Testing | \$310    | \$306    | \$314    | \$350    | \$354    | \$412    | \$292    | \$304    | \$323    | \$355    | \$351    | \$362    | \$498    | \$790    | \$659    | \$666    |
| Administration                   | \$715    | \$676    | \$710    | \$888    | \$925    | \$1,029  | \$809    | \$842    | \$1,244  | \$899    | \$826    | \$857    | \$1,010  | \$987    | \$1,061  | \$953    |
| Instructional Leadership         | \$1,015  | \$1,008  | \$1,072  | \$1,244  | \$1,257  | \$1,393  | \$1,357  | \$1,344  | \$1,431  | \$1,461  | \$1,278  | \$1,317  | \$1,318  | \$1,354  | \$1,430  | \$1,663  |
| Operations and Maintenance       | \$1,755  | \$1,427  | \$1,532  | \$1,741  | \$1,479  | \$2,421  | \$2,272  | \$1,707  | \$1,209  | \$1,765  | \$1,491  | \$1,458  | \$2,131  | \$2,296  | \$2,292  | \$2,158  |
| Other Teaching Services          | \$2,385  | \$2,040  | \$2,074  | \$2,346  | \$2,540  | \$2,352  | \$2,274  | \$2,281  | \$2,193  | \$2,175  | \$2,477  | \$2,386  | \$2,178  | \$3,119  | \$3,266  | \$3,526  |
| Pupil Services                   | \$1,085  | \$664    | \$782    | \$1,249  | \$1,259  | \$1,980  | \$1,781  | \$1,397  | \$1,759  | \$1,726  | \$1,494  | \$1,279  | \$2,192  | \$1,120  | \$1,976  | \$2,148  |
| Insurance, Retirement and Other  | \$2,098  | \$2,041  | \$2,253  | \$2,499  | \$2,753  | \$3,245  | \$3,041  | \$2,937  | \$2,943  | \$3,677  | \$3,781  | \$3,686  | \$4,766  | \$3,630  | \$3,611  | \$3,793  |
| Teachers                         | \$5,218  | \$5,690  | \$5,818  | \$6,301  | \$6,138  | \$7,547  | \$6,868  | \$6,967  | \$6,814  | \$7,967  | \$7,665  | \$7,134  | \$9,224  | \$6,888  | \$6,724  | \$6,395  |
| Total                            | \$15,156 | \$14,454 | \$15,206 | \$17,384 | \$17,321 | \$20,935 | \$19,257 | \$18,308 | \$18,512 | \$20,478 | \$19,866 | \$19,085 | \$24,358 | \$21,024 | \$21,775 | \$21,910 |

### FTE Enrollment

Table 26

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 162  | 164  | 161  | 150  | 147  | 124  | 134  | 135  | 134  | 129  | 136  | 142  | 117  | 142  | 140  | 142  |
| Foundation Enrollment      | 125  | 120  | 123  | 123  | 119  | 118  | 100  | 104  | 101  | 123  | 103  | 112  | 116  | 105  | 112  | 101  |
| Out-of-District FTE Pupils | 7    | 7    | 6    | 9    | 10   | 9    | 8    | 6    | 14   | 8    | 9    | 12   | 9    | 12   | 7    | 3    |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 27

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$2K  | \$3K  | \$3K  | \$3K  | \$2K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$4K  |
| Req NSS (minus Ch70)       | \$7K  | \$7K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$9K  | \$9K  | \$9K  |
| Actual NSS (minus Req NSS) | \$7K  | \$7K  | \$7K  | \$8K  | \$9K  | \$8K  | \$10K | \$10K | \$10K | \$8K  | \$11K | \$11K | \$10K | \$13K | \$10K | \$11K |
| Total NSS                  | \$16K | \$17K | \$17K | \$17K | \$19K | \$18K | \$21K | \$21K | \$21K | \$18K | \$23K | \$22K | \$21K | \$25K | \$22K | \$24K |

## Data: Pelham

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 28

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$580    | \$361    | \$384    | \$366    | \$658    | \$639    | \$780    | \$652    | \$569    | \$414    | \$466    | \$407    | \$292    | \$365    | \$422    | \$207    |
| Guidance, Counseling and Testing | \$438    | \$449    | \$459    | \$449    | \$617    | \$664    | \$696    | \$683    | \$682    | \$499    | \$499    | \$582    | \$795    | \$772    | \$775    | \$1,855  |
| Administration                   | \$390    | \$398    | \$478    | \$391    | \$664    | \$796    | \$806    | \$980    | \$739    | \$731    | \$489    | \$574    | \$656    | \$688    | \$614    | \$917    |
| Instructional Leadership         | \$1,123  | \$1,159  | \$1,089  | \$1,064  | \$1,113  | \$1,252  | \$1,288  | \$1,322  | \$1,278  | \$1,431  | \$1,194  | \$1,609  | \$1,670  | \$1,873  | \$1,782  | \$1,885  |
| Operations and Maintenance       | \$920    | \$847    | \$925    | \$884    | \$992    | \$947    | \$1,080  | \$942    | \$987    | \$889    | \$1,229  | \$1,304  | \$1,498  | \$1,380  | \$1,524  | \$1,614  |
| Other Teaching Services          | \$2,184  | \$2,109  | \$2,166  | \$2,026  | \$1,937  | \$2,063  | \$2,218  | \$2,541  | \$2,850  | \$2,269  | \$2,273  | \$2,324  | \$2,540  | \$2,292  | \$2,594  | \$3,010  |
| Pupil Services                   | \$594    | \$641    | \$888    | \$1,058  | \$967    | \$951    | \$1,000  | \$1,114  | \$1,092  | \$1,173  | \$1,233  | \$1,060  | \$1,317  | \$1,431  | \$1,113  | \$1,755  |
| Insurance, Retirement and Other  | \$2,615  | \$2,585  | \$2,602  | \$2,897  | \$3,052  | \$3,283  | \$3,376  | \$3,473  | \$3,863  | \$4,561  | \$3,791  | \$4,168  | \$4,850  | \$4,157  | \$3,983  | \$4,670  |
| Teachers                         | \$5,889  | \$5,474  | \$5,627  | \$5,576  | \$5,035  | \$6,090  | \$6,540  | \$6,418  | \$5,904  | \$6,310  | \$6,129  | \$6,860  | \$7,406  | \$6,714  | \$6,430  | \$7,127  |
| Total                            | \$14,733 | \$14,023 | \$14,618 | \$14,711 | \$15,035 | \$16,685 | \$17,784 | \$18,125 | \$17,964 | \$18,277 | \$17,303 | \$18,888 | \$21,024 | \$19,672 | \$19,237 | \$23,040 |

### FTE Enrollment

Table 29

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 118  | 125  | 122  | 128  | 132  | 128  | 123  | 123  | 130  | 127  | 134  | 125  | 106  | 119  | 134  | 126  |
| Foundation Enrollment      | 85   | 81   | 81   | 68   | 63   | 67   | 70   | 73   | 74   | 81   | 83   | 82   | 84   | 70   | 65   | 63   |
| Out-of-District FTE Pupils | 3    | 3    | 4    | 1    | 4    | 7    | 5    | 7    | 11   | 12   | 10   | 4    | 8    | 6    | 5    | 9    |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

Table 30

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$3K  | \$4K  | \$4K  |
| Req NSS (minus Ch70)       | \$5K  | \$6K  | \$6K  | \$7K  | \$8K  | \$8K  | \$9K  | \$9K  | \$8K  | \$8K  | \$8K  | \$8K  | \$8K  | \$9K  | \$10K | \$10K |
| Actual NSS (minus Req NSS) | \$8K  | \$8K  | \$8K  | \$10K | \$12K | \$10K | \$9K  | \$10K | \$12K | \$12K | \$11K | \$11K | \$13K | \$16K | \$19K | \$20K |
| Total NSS                  | \$16K | \$17K | \$17K | \$20K | \$23K | \$22K | \$21K | \$22K | \$23K | \$23K | \$22K | \$23K | \$24K | \$29K | \$32K | \$34K |

## Data: Shutesbury

PPE (\$/in-district FTE pupil), FTE Enrollment, and NSS/Ch70 Funding (\$/foundation enrollment pupil)

### PPE by Category (\$/in-district FTE pupil)

Table 31

| Category                         | 2009     | 2010     | 2011     | 2012     | 2013     | 2014     | 2015     | 2016     | 2017     | 2018     | 2019     | 2020     | 2021     | 2022     | 2023     | 2024     |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Other                            | \$580    | \$498    | \$564    | \$865    | \$704    | \$515    | \$636    | \$684    | \$1,510  | \$420    | \$385    | \$873    | \$1,296  | \$551    | \$1,102  | \$545    |
| Guidance, Counseling and Testing | \$302    | \$328    | \$345    | \$334    | \$345    | \$344    | \$354    | \$434    | \$318    | \$546    | \$542    | \$383    | \$519    | \$1,393  | \$697    | \$826    |
| Administration                   | \$652    | \$943    | \$767    | \$836    | \$877    | \$726    | \$884    | \$1,084  | \$1,040  | \$846    | \$856    | \$938    | \$925    | \$1,100  | \$934    | \$1,015  |
| Instructional Leadership         | \$969    | \$1,138  | \$1,149  | \$1,198  | \$1,227  | \$1,003  | \$1,192  | \$1,517  | \$1,604  | \$1,481  | \$1,517  | \$1,541  | \$1,707  | \$1,736  | \$1,323  | \$2,103  |
| Operations and Maintenance       | \$1,180  | \$1,079  | \$1,221  | \$1,276  | \$1,251  | \$1,338  | \$1,384  | \$1,637  | \$1,791  | \$1,941  | \$1,663  | \$1,916  | \$2,665  | \$2,048  | \$2,051  | \$2,754  |
| Other Teaching Services          | \$2,185  | \$2,447  | \$2,361  | \$2,193  | \$2,179  | \$2,591  | \$2,164  | \$2,817  | \$3,301  | \$3,491  | \$3,545  | \$3,560  | \$3,722  | \$5,320  | \$3,441  | \$4,359  |
| Pupil Services                   | \$990    | \$800    | \$850    | \$1,222  | \$1,283  | \$1,074  | \$1,073  | \$906    | \$1,847  | \$1,755  | \$1,664  | \$1,815  | \$1,937  | \$1,355  | \$2,249  | \$2,136  |
| Insurance, Retirement and Other  | \$2,332  | \$2,687  | \$2,596  | \$2,690  | \$2,564  | \$2,484  | \$2,328  | \$3,107  | \$3,773  | \$4,388  | \$2,970  | \$3,963  | \$3,350  | \$3,154  | \$4,007  | \$3,501  |
| Teachers                         | \$4,821  | \$5,363  | \$5,603  | \$5,169  | \$5,509  | \$6,012  | \$5,620  | \$6,612  | \$7,043  | \$7,838  | \$7,914  | \$8,204  | \$8,933  | \$10,075 | \$8,898  | \$9,669  |
| Total                            | \$14,011 | \$15,283 | \$15,456 | \$15,783 | \$15,939 | \$16,087 | \$15,635 | \$18,798 | \$22,227 | \$22,706 | \$21,056 | \$23,193 | \$25,054 | \$26,732 | \$24,702 | \$26,908 |

### FTE Enrollment

Table 32

| FTE Series                 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|----------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| In-District FTE Pupils     | 162  | 151  | 146  | 151  | 150  | 157  | 156  | 131  | 120  | 124  | 127  | 125  | 113  | 115  | 123  | 110  |
| Foundation Enrollment      | 156  | 149  | 141  | 138  | 143  | 132  | 141  | 146  | 118  | 116  | 118  | 117  | 105  | 98   | 100  | 111  |
| Out-of-District FTE Pupils | 7    | 5    | 7    | 10   | 9    | 13   | 9    | 12   | 13   | 16   | 18   | 15   | 16   | 18   | 19   | 18   |

### NSS/Ch70 Funding Components (\$/foundation enrollment pupil)

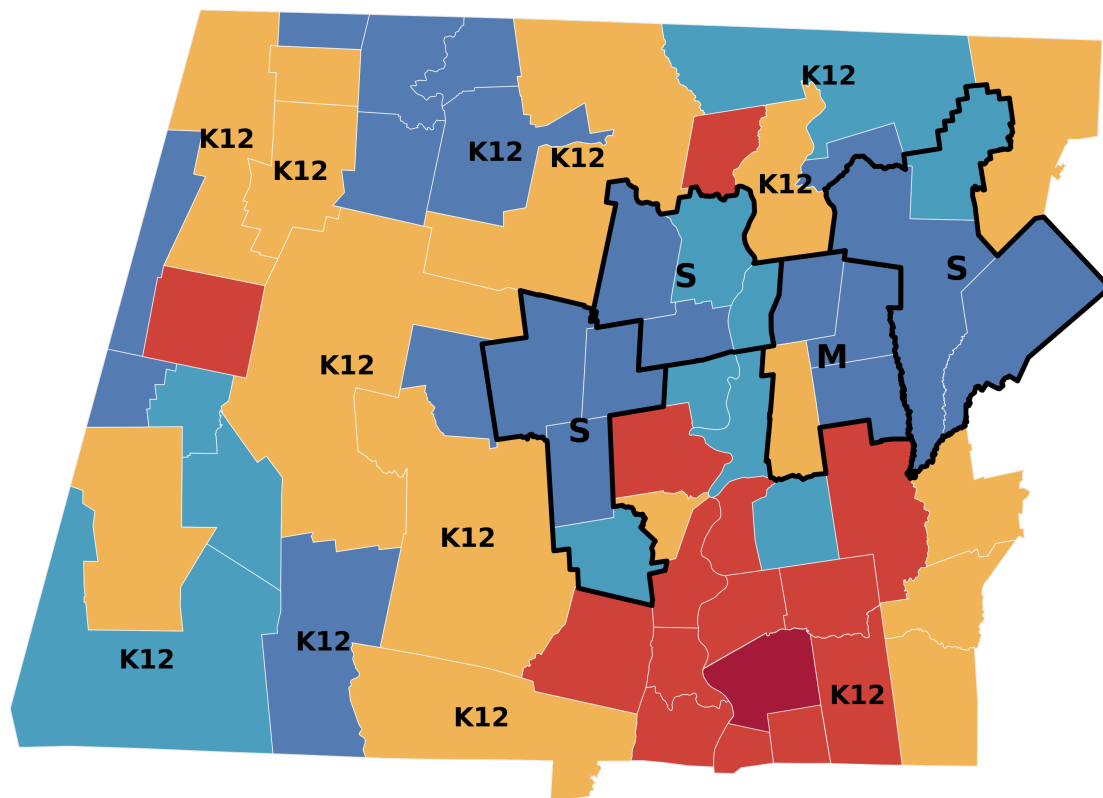
Table 33

| Component                  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Ch70 Aid                   | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$6K  |
| Req NSS (minus Ch70)       | \$4K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$5K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$7K  | \$7K  |
| Actual NSS (minus Req NSS) | \$4K  | \$5K  | \$6K  | \$6K  | \$6K  | \$7K  | \$7K  | \$6K  | \$9K  | \$10K | \$10K | \$12K | \$12K | \$15K | \$13K | \$10K |
| Total NSS                  | \$13K | \$14K | \$15K | \$15K | \$15K | \$17K | \$16K | \$15K | \$20K | \$22K | \$21K | \$23K | \$25K | \$29K | \$26K | \$24K |



## Appendix C. Maps from previous years

Geographic map showing district locations and enrollment cohorts (2019)



Enrollment Cohorts - 2019

|                                                |                                                                |
|------------------------------------------------|----------------------------------------------------------------|
| Tiny (0-200 FTE): 20 district(s)               | Secondary regional (black border)                              |
| Small (201-800 FTE): 14 district(s)            | Cohort: T, S, M, L, XL                                         |
| Medium (801-1600 FTE): 16 district(s)          | K12 = Regional K-12 with all grade levels in a single district |
| Large (1601-10K FTE): 14 district(s)           |                                                                |
| Outliers (Springfield >10K FTE): 1 district(s) |                                                                |

Figure 1

This appendix contains historical geographic maps showing the evolution of district enrollment cohorts and per-pupil expenditures over time (2009, 2014, 2019). These visualizations provide context for understanding how enrollment patterns and spending levels have shifted across Western Massachusetts districts.

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

## Appendix C. Maps from previous years (continued)

Geographic map showing 2019 PPE vs enrollment cohort baseline

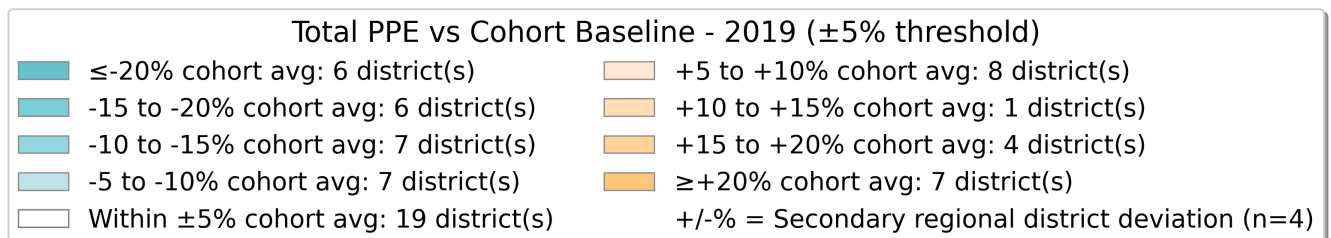
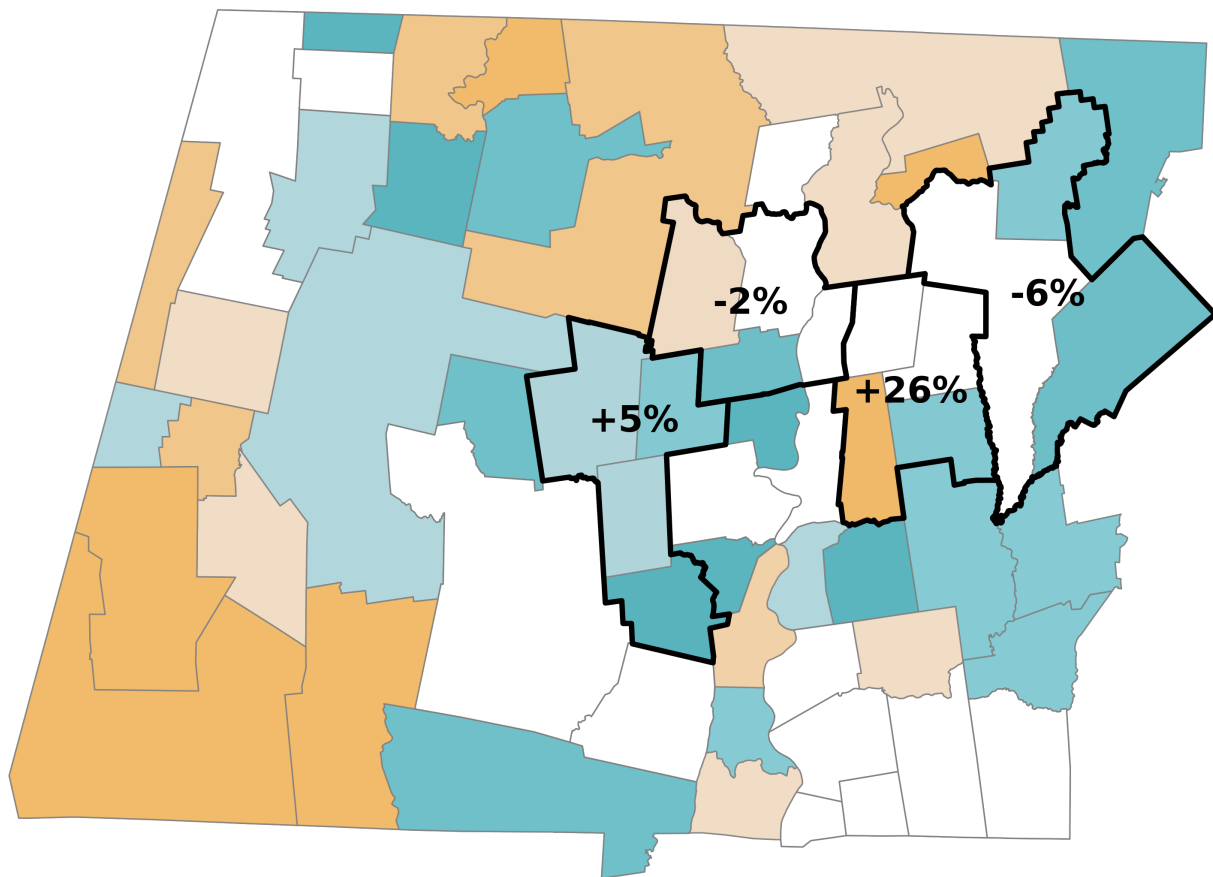


Figure 2

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort). This helps identify which districts are operating above or below the typical spending level for their enrollment size, accounting for the well-documented relationship between district size and per-pupil costs.

## Appendix C. Maps from previous years (continued)

Geographic map showing 10-year PPE growth (2009-2019) vs enrollment cohort baseline

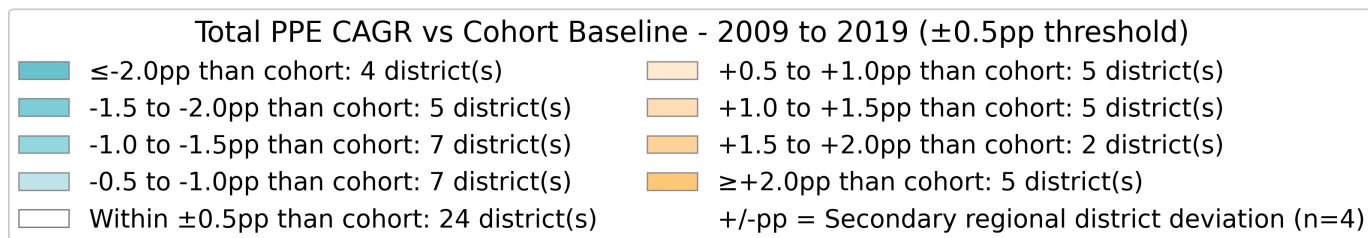
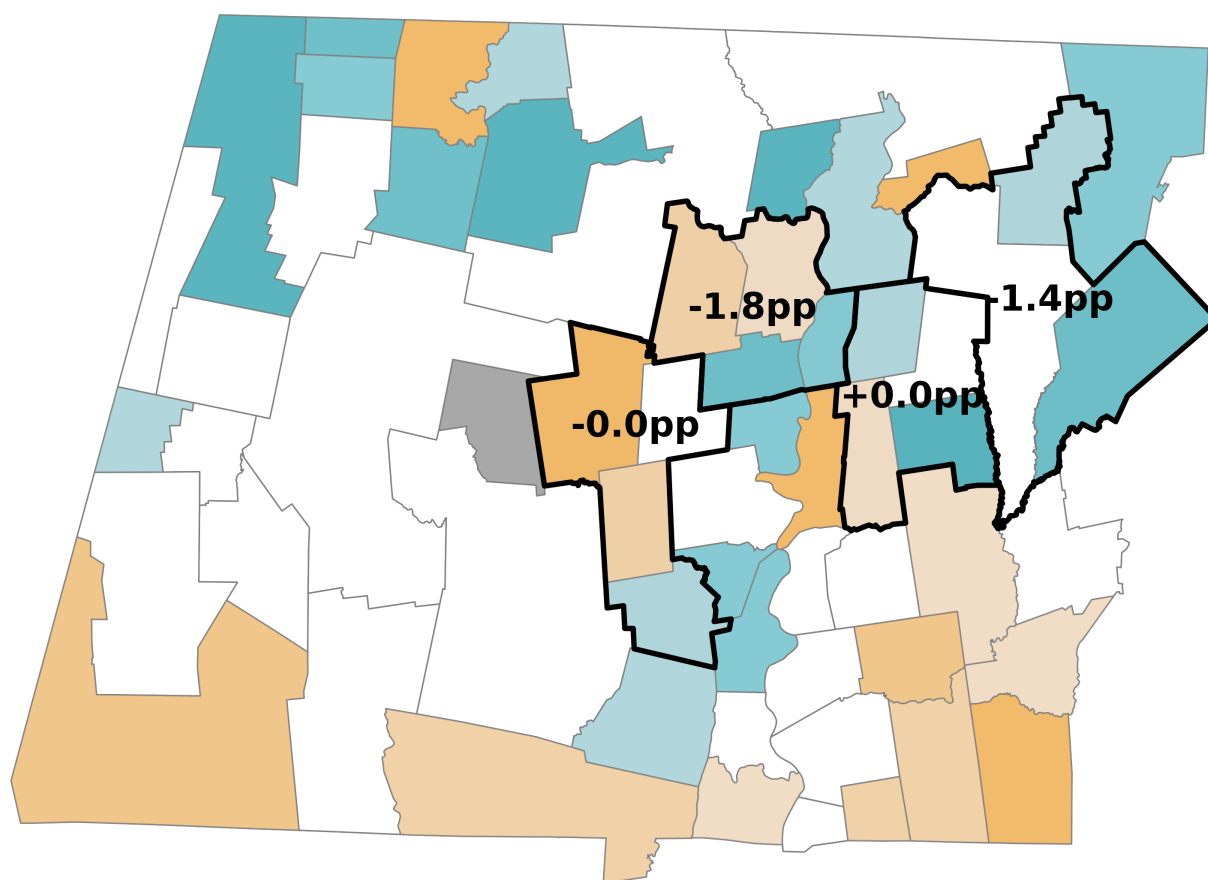


Figure 3

This map shows each district's 15-year spending growth rate (CAGR, 2009-2024) compared to its enrollment cohort baseline growth rate. This helps identify which districts experienced faster or slower spending growth than similar-sized districts, revealing local policy decisions and cost pressures beyond typical enrollment cohort trends.

## Appendix C. Maps from previous years (continued)

Geographic map showing district locations and enrollment cohorts (2014)

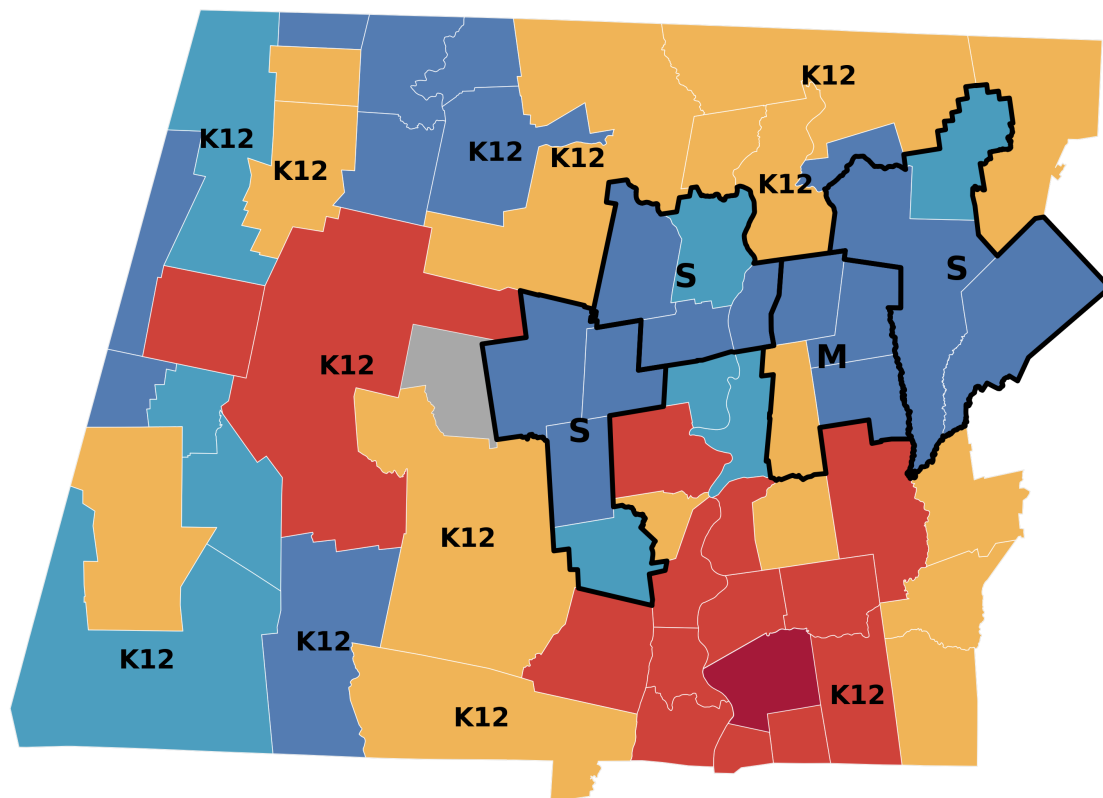


Figure 4

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

## Appendix C. Maps from previous years (continued)

Geographic map showing 2014 PPE vs enrollment cohort baseline

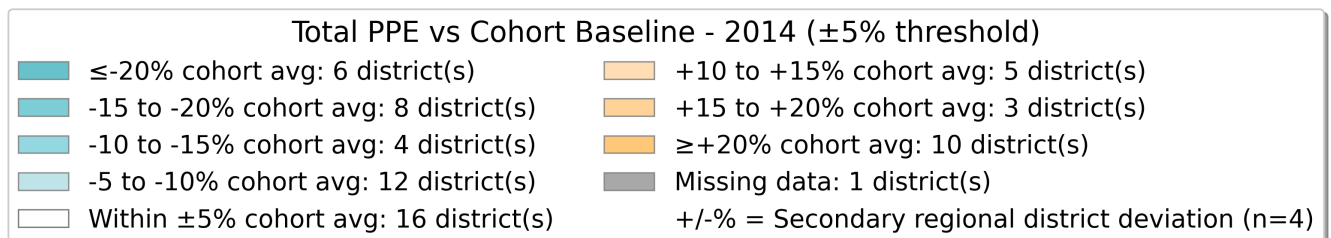
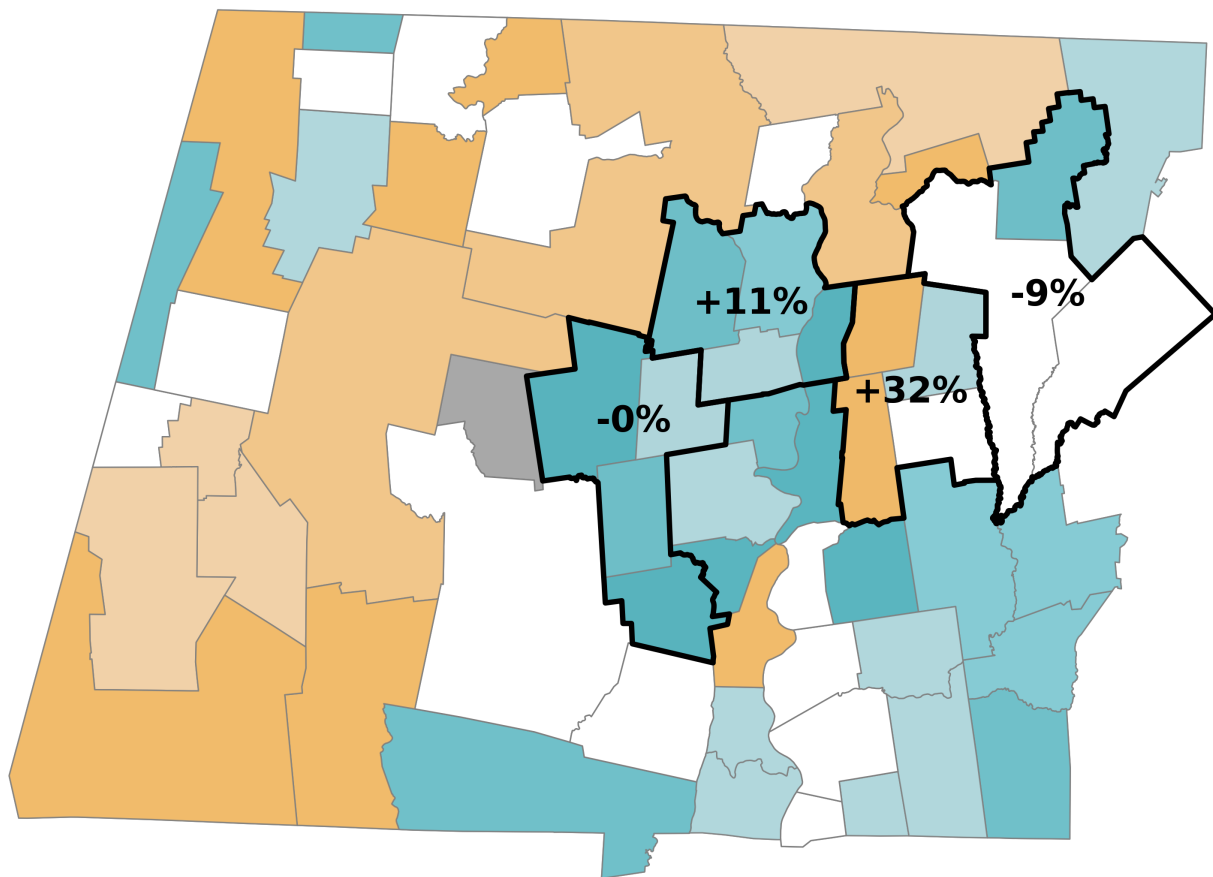


Figure 5

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort). This helps identify which districts are operating above or below the typical spending level for their enrollment size, accounting for the well-documented relationship between district size and per-pupil costs.

## Appendix C. Maps from previous years (continued)

Geographic map showing 5-year PPE growth (2009-2014) vs enrollment cohort baseline

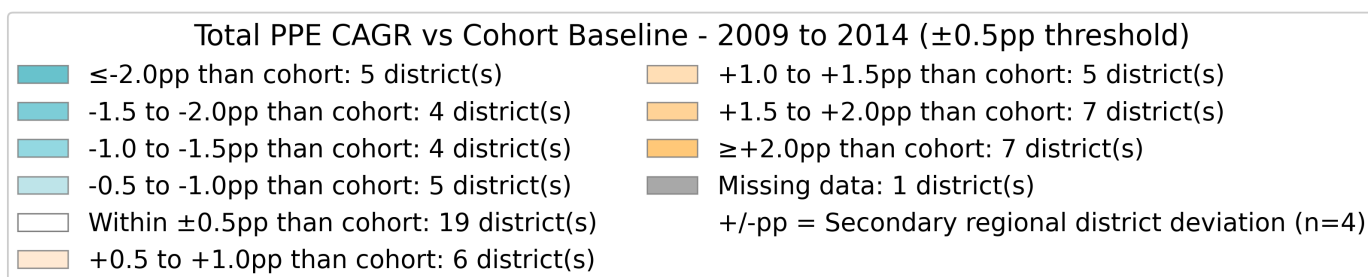
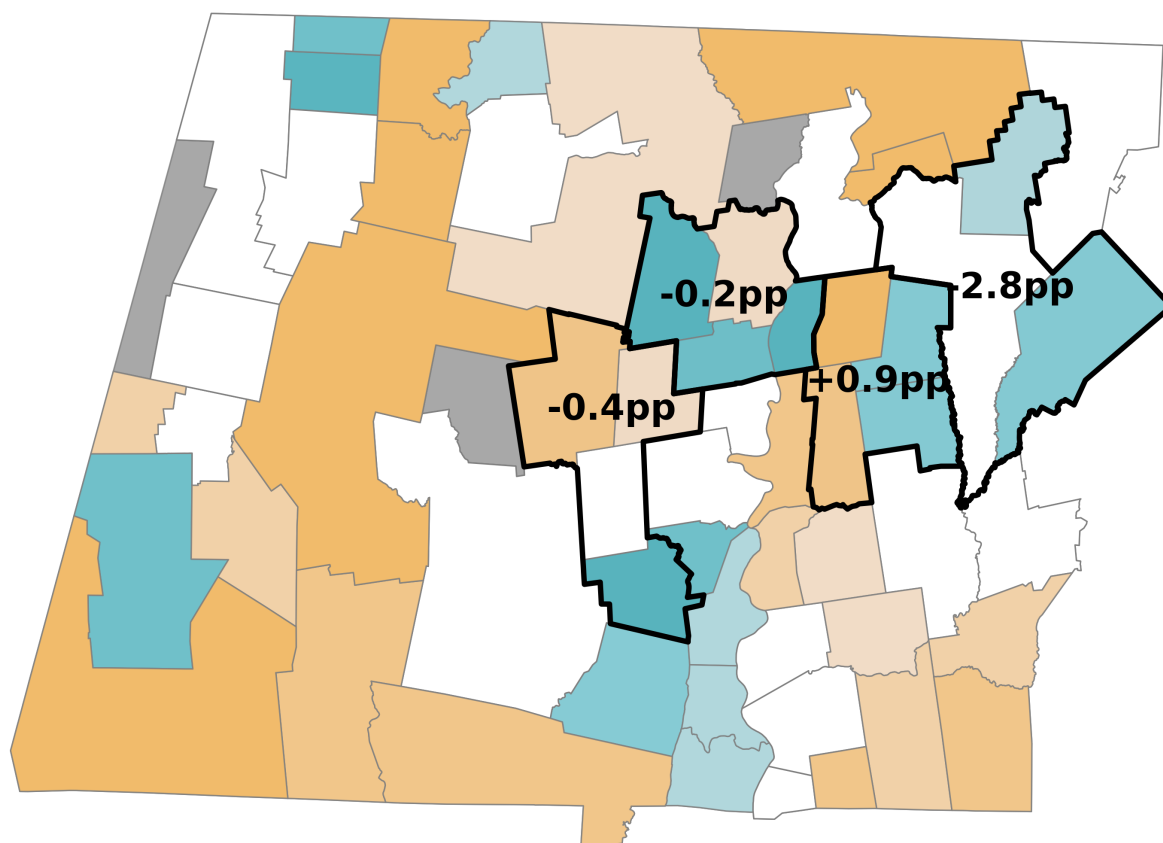


Figure 6

This map shows each district's 15-year spending growth rate (CAGR, 2009-2024) compared to its enrollment cohort baseline growth rate. This helps identify which districts experienced faster or slower spending growth than similar-sized districts, revealing local policy decisions and cost pressures beyond typical enrollment cohort trends.

## Appendix C. Maps from previous years (continued)

Geographic map showing district locations and enrollment cohorts (2009)

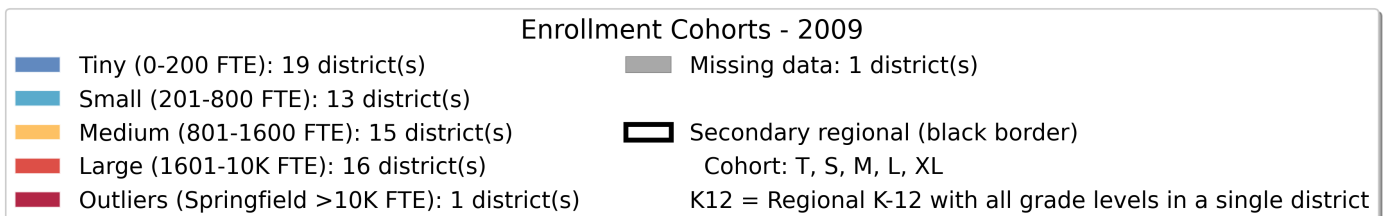
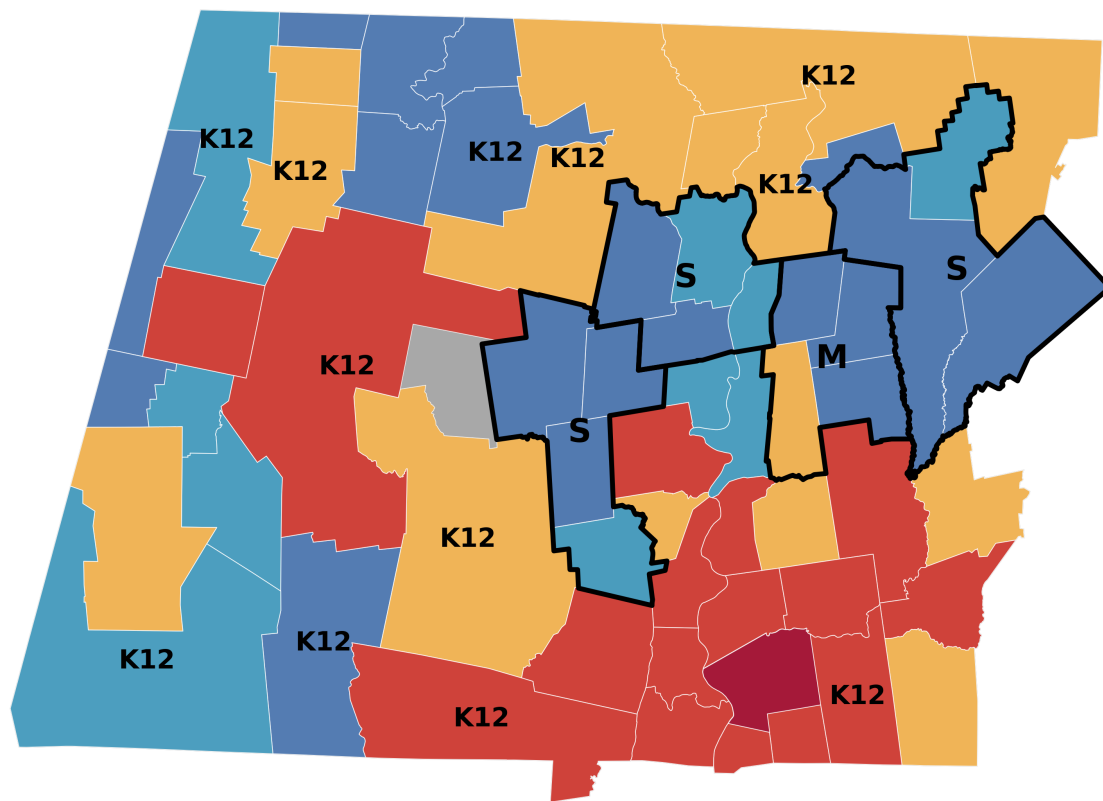
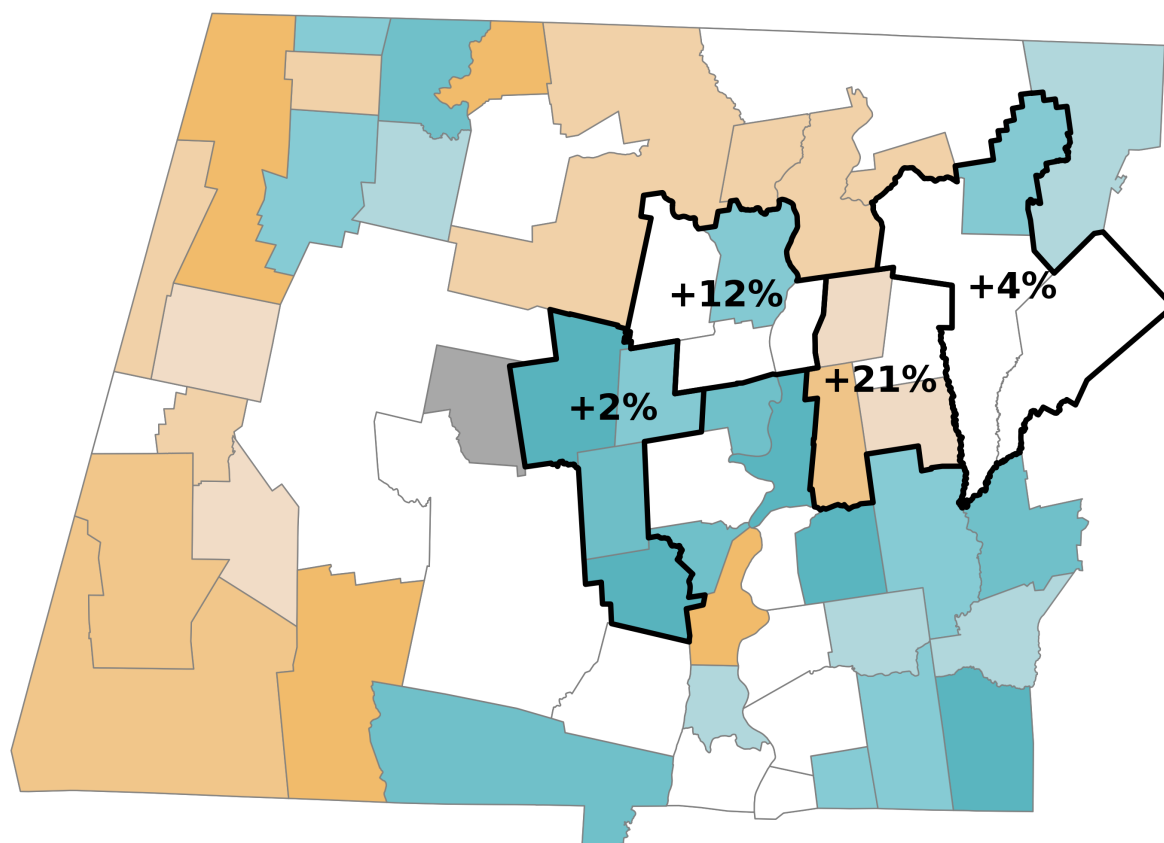


Figure 7

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

## Appendix C. Maps from previous years (continued)

Geographic map showing 2009 PPE vs enrollment cohort baseline



### Total PPE vs Cohort Baseline - 2009 (±5% threshold)

|                                       |                                                    |
|---------------------------------------|----------------------------------------------------|
| ≤-20% cohort avg: 5 district(s)       | +10 to +15% cohort avg: 8 district(s)              |
| -15 to -20% cohort avg: 6 district(s) | +15 to +20% cohort avg: 3 district(s)              |
| -10 to -15% cohort avg: 8 district(s) | ≥+20% cohort avg: 5 district(s)                    |
| -5 to -10% cohort avg: 5 district(s)  | Missing data: 1 district(s)                        |
| Within ±5% cohort avg: 20 district(s) | +/-% = Secondary regional district deviation (n=4) |
| +5 to +10% cohort avg: 4 district(s)  |                                                    |

Figure 8

This map shows each district's 2024 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort). This helps identify which districts are operating above or below the typical spending level for their enrollment size, accounting for the well-documented relationship between district size and per-pupil costs.